

Session 17: Correlation & Simple Regression — Pen-and-Paper Pair Exercise

PSY 410 | Data Science for Psychology

Name: _____ Date: _____

No laptop today? No problem. Work with a partner who has one — you do the paper version, they run the code, you compare at the end.

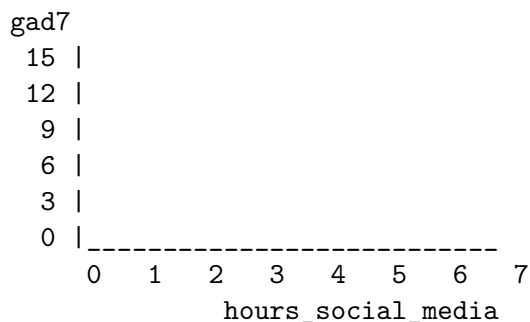
The data

Eight students, daily hours of social media use, and GAD-7 anxiety score:

student	hours	gad7	student	hours	gad7
1	2.1	5	5	6.2	14
2	5.3	12	6	4.5	8
3	1.0	3	7	0.5	4
4	3.8	9	8	3.0	7

Step 1: Sketch a scatterplot

Plot the 8 points on the grid.



Pattern: positive / negative / none? Strong / moderate / weak?

Your read: _____

Step 2: Predict r

Based on the shape, circle your best guess:

~0 • 0.3 • 0.6 • 0.9 • -0.6 • -0.9

Step 3: Read your partner's output

Your partner runs `lm(gad7_score ~ hours_social_media, data = student_data)` and `summary(model)`. Record what they see:

Component	Value	What it means
Intercept (b0)	_____	Predicted GAD-7 when hours = 0

Component	Value	What it means
Slope (b1)	_____	Change in GAD-7 per +1 hour of social media
R-squared	_____	Proportion of variation in GAD-7 <i>predictable from</i> hours
p (slope)	_____	Significant at $p < .05$? Y / N

Step 4: Make a prediction

Using b0 and b1 from Step 3, predict the GAD-7 score for a student who uses social media 4 hours per day:

Predicted = _____ + _____ $\times$ 4 = _____

Step 5: Reconstruct the code

Fill in the blanks from memory before checking your partner's screen:

```
cor(student_data$_____, student_data$_____)

model <- lm(_____ ~ _____, data = student_data)
summary(model)

ggplot(student_data, aes(x = _____, y = _____)) +
  geom_point() +
  geom_smooth(method = "____") +
  labs(title = "_____",
       x = "_____", y = "_____")
```

Step 6: Correlation vs. causation

Does this analysis prove that social media *causes* anxiety? Why or why not? Name at least one alternative explanation for the pattern.

Done? Compare your sketch, your r prediction, and your code with your partner's screen. Do the numbers match what you expected?